

Functions

Functions are used apply certain operations to the data.



1. Single-row functions()

Input: **1** value

Output: **1** value

Example: LOWER()

2. Multi-row Functions()

Input: **Multiple** values

Output: **1** value

Example: SUM()

Nesting Functions

Chaining functions back to back

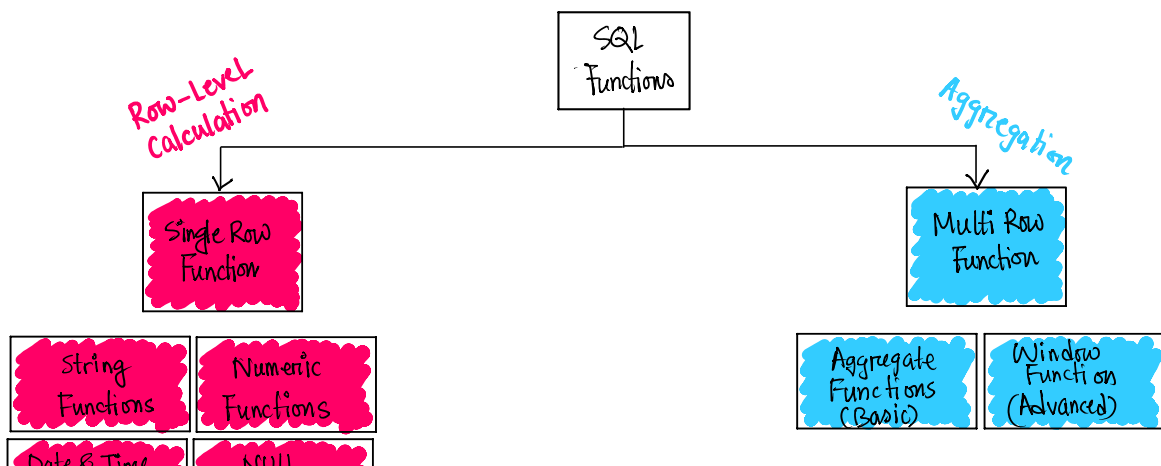
Example:

1. 'Maria' → LEFT(2) → 'Ma' → LOWER() → 'ma'

→ LOWER(LEFT('Maria', 2))

2. LEN(LOWER(LEFT('Maria', 2)))

Types



Functions	Functions
Date & Time Functions	NULL Functions

Functions (Basic)	Functions (Advanced)
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(Data engineering)

(Data Analysis)

STRING FUNCTIONS

Manipulation

- CONCAT
- UPPER
- LOWER
- TRIM
- REPLACE

Calculation

- LEN

String Extraction

- LEFT
- RIGHT
- SUBSTRING

CONCAT

Concatenate strings

UPPER

All characters to upper case

LOWER

All characters to lower case

TRIM

Remove leading and trailing spaces.

REPLACE

Replace specific character with new character.

Say we have 123-456-7890
Let's replace '-' with '/'

REPLACE

old-value = '-'

new-value = '/' → To add blank space,
new-value = ' ' → To add empty string

```
SELECT
  col-1
  , LEN(col-1)
  , LEN(TRIM(col-1))
FROM Table-1
WHERE col-1 != TRIM(col-1)
WHERE LEN(col-1) != LEN(TRIM(col-1))
```

old-value = '—'

new-value = '/' → To add blank space,
It can be left empty: ''

SYNTAX

REPLACE('base-string', <old-value>, <new-value>)

LEN

Number of characters in the given string

Takes any data type

LEFT

Get specific number of character from the start

Syntax: LEFT('string-value', num-char)

RIGHT

Get specific number of character from the end

Syntax: RIGHT('string-value', num-char)

SELECT

col-name,

LEFT(TRIM(col-name), 2),

RIGHT(col-name, 2)

FROM Table-name

SUBSTRING

Extract part of a string from a specified position

Syntax:

SUBSTRING(<source-string>, <start position>, <length>)

→ For cases
where you need
rest of the
characters, you
can use the
LEN() function.